

ALEXEY GALLEGO

Solutions Engineer | Full-Stack Development | Systems & Automation | AI Integration
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TECHNICAL SKILLS

Languages & Frameworks: JavaScript (Node.js, React, Angular), Python, Java, Spring Boot, Next.js, REST APIs, ASP.NET

AI & Automation: LLM Engineering, Agentic AI, MCP, OpenAI / Claude APIs, Codex / Claude Development

Databases: MSSQL, PostgreSQL, MySQL, MongoDB, Sequelize ORM

Infrastructure & DevOps: Docker, Kubernetes, Linux, CI/CD, GitHub Actions, AWS, VMware

Engineering: FANUC Robot Programming, PLC, SolidWorks, 3D Printing, KiCAD

Methodologies: Agile/SCRUM, MVC Architecture, JWT/OAuth2.0

EXPERIENCE

Systems & Software Engineer | Encore Automation | Auburn Hills, MI *Jul 2021 – Present*

- Leading a team of 3 engineers in designing and building a full enterprise ERP system (ASP.NET, Angular) covering timesheets, expenses, inventory, parts management, requisitions, contacts, organizations, sales, engineering releases, IT ticketing, and heavy reporting, driving architecture decisions and full stack development across the full system.
- Individually architected and delivered an internal business operations platform from scratch (Node.js, React) serving 110 employees, replacing manual spreadsheet workflows with automated timesheet entry, expense management, event scheduling, deadline tracking, and multi-level approval routing.
- Eliminated manual administrative follow-up by building automated email reminder systems that alert employees to missing or past-due timesheets and provide managers a real-time dashboard of outstanding submissions across the organization.
- Maintained 99%+ uptime across all internal platforms by owning production infrastructure on self-hosted Ubuntu servers, including Docker deployments, CI/CD pipelines, and reverse proxy configuration.
- Secured internal platforms end-to-end by implementing JWT/refresh-token authentication with rotating sessions, CSRF protection, and role-based access control across multi-roles scopes.
- Improved industrial sealing process reliability from ~80% to 99% by building a Python + Tkinter application that interfaced directly with FANUC robots via serial connection to extract real-time pressure sensor data, flag faulty components, and log results to a structured database.
- Rebuilt the company's web presence from the ground up using Next.js, replacing an outdated site with outdated product information, and extended brand reach by producing and editing video content for LinkedIn and YouTube.

PROJECTS

AI Receipt Analysis System | Solo Developer 2026 – Present

- Built an AI pipeline (OpenAI Vision API) that analyzes receipt images, extracts structured line-item data, and matches entries to expense records, integrated as a core module of the ERP platform reducing manual review time for administrators.

FANUC Robot Data Pipeline | Solo Developer 2022

- Built a Python GUI application that connected to FANUC industrial robots via serial interface, extracted pressure sensor telemetry, flagged defective components, and stored results in a database, raising process reliability from around 80% to 99%.

Makerspace Operations Platform | Co-Founder / Engineer 2025 – Present

- Co-building a platform for makerspace operations management covering member workflows, resource scheduling, and operational tooling, using React, Node.js, and Codex spec driven development.

EDUCATION

B.S. Mechanical Engineering | Florida International University, Miami, FL | May 2021